

LNP's and Neurology

The nervous system is a system of electrochemical signaling. Neurological conditions that develop, disrupt this signaling system and lead to severe and dire consequences for patients. Damage to this system—depending on what condition it is—can originate in tissues, neurons, or proteins (if they are misfolded). Such conditions are even harder to treat because mature neurons cannot divide, so the human body cannot regenerate the neurons that it has lost. Along with this, the neurons are hyper-sensitive to stress and fluctuations in the microenvironment.

Even when researchers can figure out which protein to be replaced or treated, it is very hard to deliver the right drug to the brain. This is because of the blood-brain barrier—a layer of cells that acts like a door, letting in and letting out only certain types of molecules or ions—most of which don't include the treatment being given. Most drugs are redirected to the liver and spleen, never reaching the brain in the first place.

Along with this, a lot of the early screening and testing for these therapies is conducted in a lab-raised cell culture rather than the actual body. This means that the drugs are tested in conditions that do not completely replicate the complexity of the nervous system. Therefore, a treatment can seem effective in vitro and fail once introduced into the body. Due to these challenges, researchers have been exploring new methods to deliver treatments more effectively, including lipid nanoparticle (LNP) transfection.

Lipid nanoparticles are tiny fat bubbles that carry mRNA, the instructions for protein synthesis, into cells through a form of transfection, such as an injection. Even though LNPs can help with delivery, they can still interact with the immune system once they enter the body. However, the immune system may recognize the LNP, the mRNA, or the protein as unfamiliar and respond by letting out an alarm, in the form of cytokines or chemokines, calling other immune cells for help. This strong immune response can lead to temporary symptoms such as headache, fatigue, or brain fog.

In rare cases, certain neurological conditions have been reported and investigated, such as Bell's palsy (which affects the facial nerve), Guillain-Barré syndrome (which occurs when the immune system attacks peripheral nerves), and transverse myelitis (spinal cord inflammation). One possible explanation is an unusually strong or misdirected immune response, which may increase inflammation or affect the blood-brain barrier (BBB). Scientists are also studying autoimmunity, where the immune system accidentally attacks the body's own cells. One proposed mechanism, called molecular mimicry, suggests that a foreign protein may resemble a human protein closely enough that the immune response occasionally reacts against both.

Nevertheless, correlation doesn't mean causation, as the relationship between the vaccine and common symptoms is still being researched. Aside from studying possible immune effects, researchers are also trying to solve another major problem: getting treatments across the blood-brain barrier.

The blood-brain barrier is a highly selective barrier formed mainly by tightly connected endothelial cells lining blood vessels in the brain. To overcome this major obstacle in treating central nervous system disorders, researchers are developing innovative strategies that will allow

lipid nanoparticles to be able to safely pass through without surgery (as brain surgery can often be very dangerous and expensive).

Cells in your body have natural receptors on their outer surface which the brain uses to pull in essential nutrients it actually needs. Targeting molecules can bind to specific receptors on blood-brain barrier cells and help transport therapeutic cargo across the barrier.

Lipid formulations are recipes that contain the many different fats and oils that make up the outer wall of the nanoparticle. If this recipe is wrong, the cell will either block it or be destroyed by it. Through much trial and error, researchers found when inserting one with a specific lipid called DOPE, it bends and naturally fuses right into the cell's outer skin. Due to these properties, medicine can be safely delivered without damaging the cell or tissue.

Usually when medicine is taken as a pill or regular shot, it enters your bloodstream which carries it through your whole body, passing through the heart, liver, and spleen first, and by the time it reaches the brain the blood-brain barrier stops almost all of it from entering. Intranasal delivery may use olfactory and trigeminal pathways to improve transport toward the brain, although delivery is not completely efficient.

Another strategy researchers are exploring is focused ultrasound, which can temporarily make the blood-brain barrier easier to cross. The main barrier comes from tight junctions between endothelial cells in brain blood vessels, not rigid brain cell walls; however, exposing them to low frequency sound waves causes extremely gentle, invisible vibrations in the cell walls, which shake the cell surface just enough to make it loose, without damaging the cell. This allows the cell to open up and absorb more medicine genes than they normally would.

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